Guide: Using Merge Rules After an Orbit Shift

Collatz proof project — September 5, 2026

What changed

Earlier rules could replace certain hypothetical noncontracting seeds with smaller ones. A noncontracting seed is one whose multiplicative coefficient never falls below one. Moving forward along its orbit can lose that property, so applying the old rule theorem directly to a shifted value would leave a gap.

The new Lean proof keeps a numerical lower bound on every future coefficient. A rule improves that bound. Several rules can restore it to one, at which point the resulting seed is fully noncontracting again. If that seed is smaller than the original, the original could not have been the least positive noncontracting seed.

How the accounting works

Represent the lower bound as $1/D$, with $D \geq 1$. Moving forward by k steps from a fully noncontracting seed supplies the initial value $D = C_k$, the coefficient accumulated during that shift. A merging rule with coefficient boost ρ updates the bound by

$$D \longmapsto \max(1, D/\rho).$$

The rule must have an actual merging path and a noncontracting replacement prefix. These hypotheses are checked; they are not assumed for arbitrary paths. All calculations use exact integers or fractions.

A verified infinite family

For every natural number Q, Lean proves the conditional implication

$$447 + 549755813888Q \text{ is noncontracting} \quad \Longrightarrow \quad 307 + 376572715308Q \text{ is noncontracting.}$$

The second seed is always positive and smaller. At $Q = 0$, the proof shifts 447 forward 23 steps to 767, then applies three rules producing 511, 461 and 307. The intermediate deficits exceed one; the third rule restores the bound to one. The theorem applies to every quotient, not just these four displayed small seeds.

What the experiment says

Of 2,336 finite seeds left unresolved by the earlier search, the new search finds chain certificates for 1,012. For 182 of these, no single rule suffices at any searched shift. Another 1,216 reach coefficient contraction before a chain is found; 108 remain at the 64-step horizon. These counts concern the actual finite seeds, not all numbers sharing their initial parity patterns. No resource cap was reached. Independent orbit replay checks every saved chain, but the full census is not a Lean completeness theorem.

What still needs to be proved

We need an argument that a suitable chain exists for every hypothetical least positive noncontracting seed. Neither the finite search nor the infinite family supplies that coverage. The nontrivial-cycle problem also remains. Collatz is not proved.

Where to check the work

The formal proofs are in `lean/Collatz/MergeDeficit.lean`. From `lean/`, run `lake build Collatz.MergeDeficit`. From the root, run `python3 analysis/merge_chain_search.py` to reproduce the census, or `python3 -m unittest discover -s analysis -p test_merge_chain_search.py` for independent checks. The technical paper gives the exact hypotheses and family arithmetic.